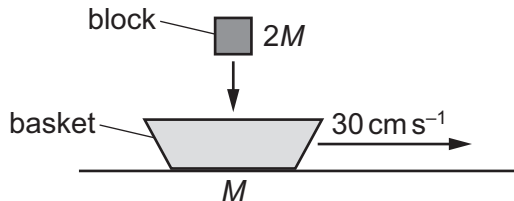


- 12** A basket is sliding on a horizontal frictionless surface with a velocity of 30 cm s^{-1} .



The mass of the basket is M .

A block of mass $2M$ falls vertically into the basket. The block and the basket then move together with the same horizontal speed v .

What is v ?

- A** 10 cm s^{-1} **B** 15 cm s^{-1} **C** 30 cm s^{-1} **D** 60 cm s^{-1}